# Task-1 on git and github

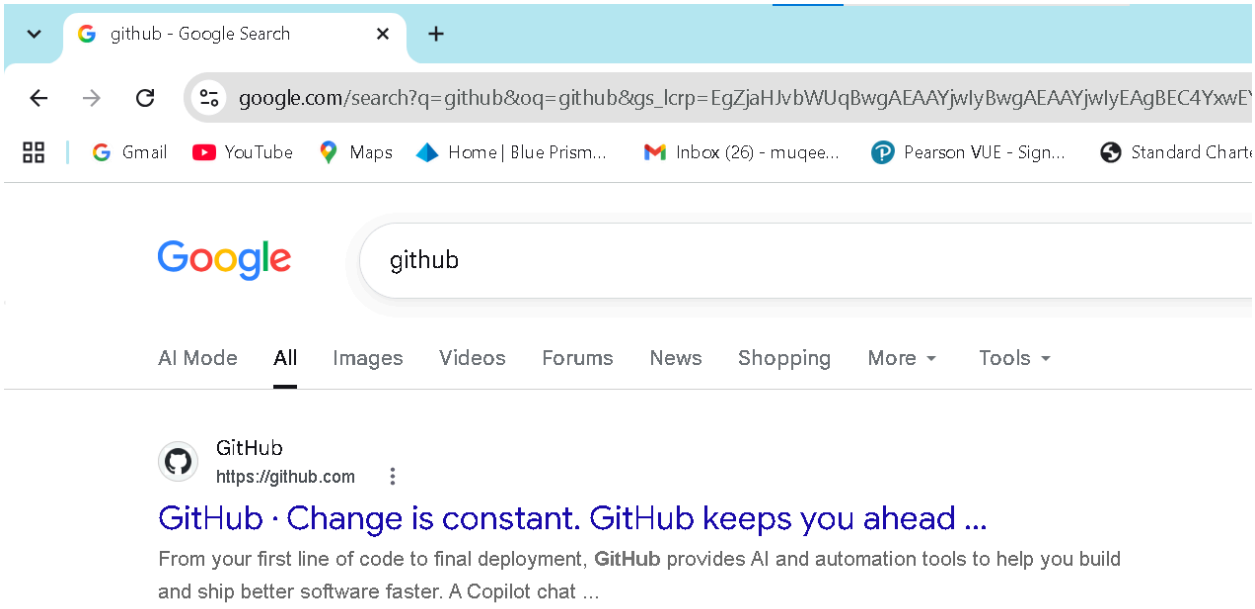
**Watch the video and write notes.**

Watching the video has been completed.

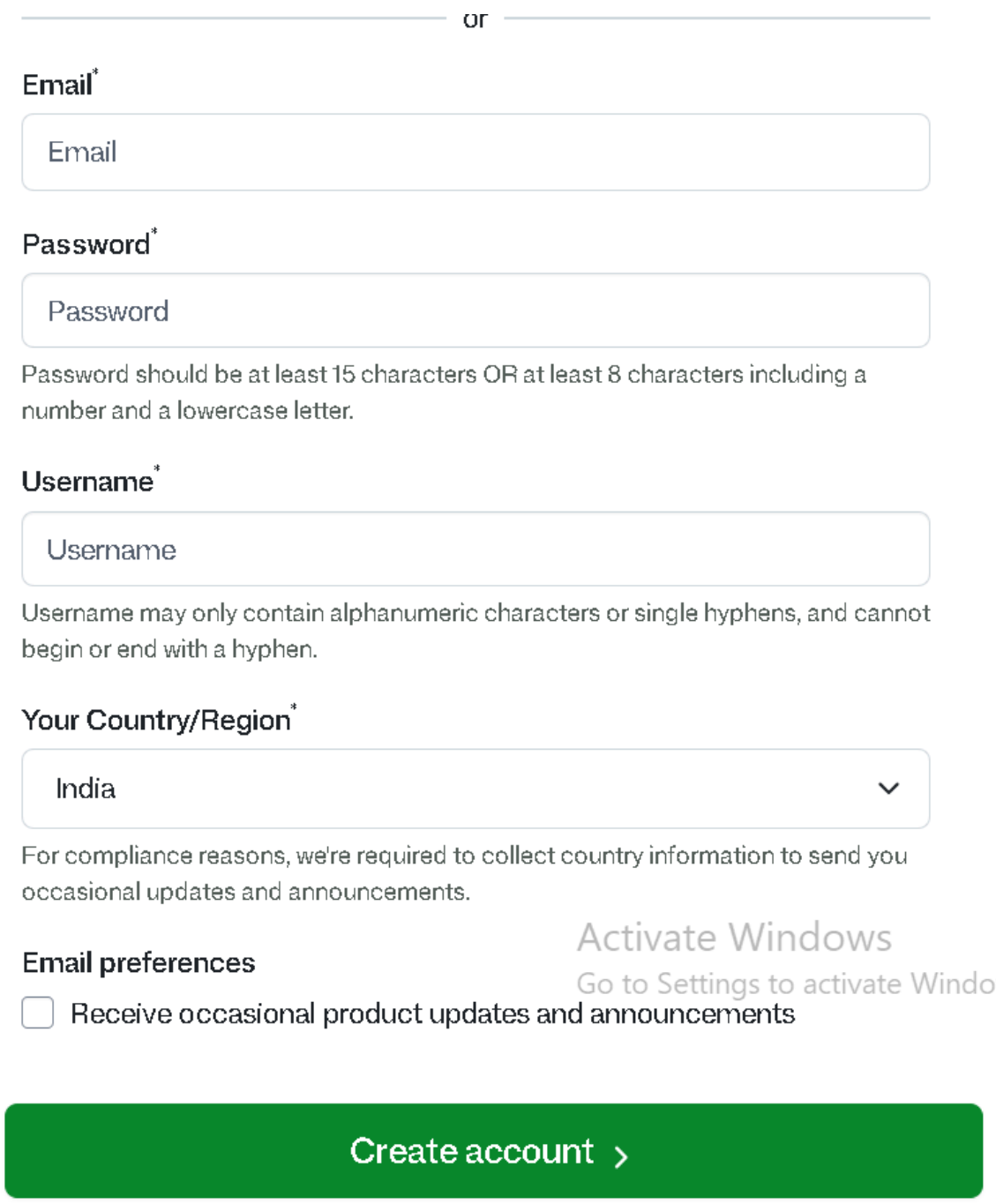
Prepared the notes

**Create a GitHub account.**

Step1: Open Github in your browser and click the first link




Step2: Click signup and fill the details and click on create account.




Step 3: Verify your account

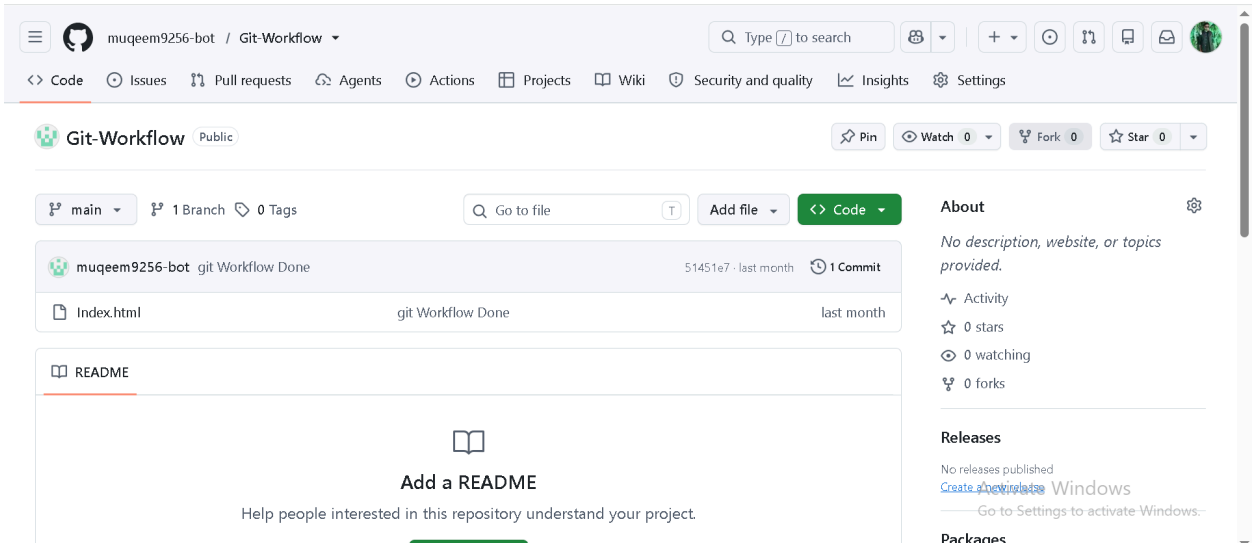
Complete the verification puzzle if requested. Follow the instructions to create your account.

Step 4: Verify your email

Check your email inbox for the verification code. Enter it on GitHub when prompted.

Step 5: Start using GitHub

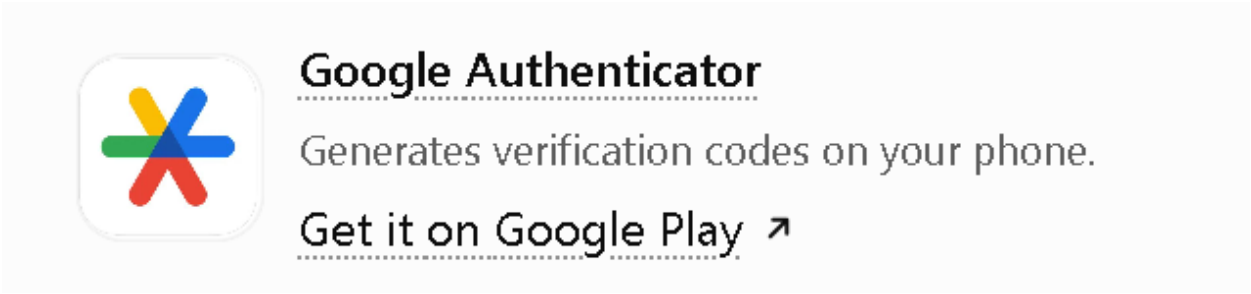
Once your account is created, you can sign in and create your first repository to store your DevOps scripts.




**Set up MFA (Multi-Factor Authentication) for the GitHub account using a mobile app.**

## Step 1: Install an authenticator app

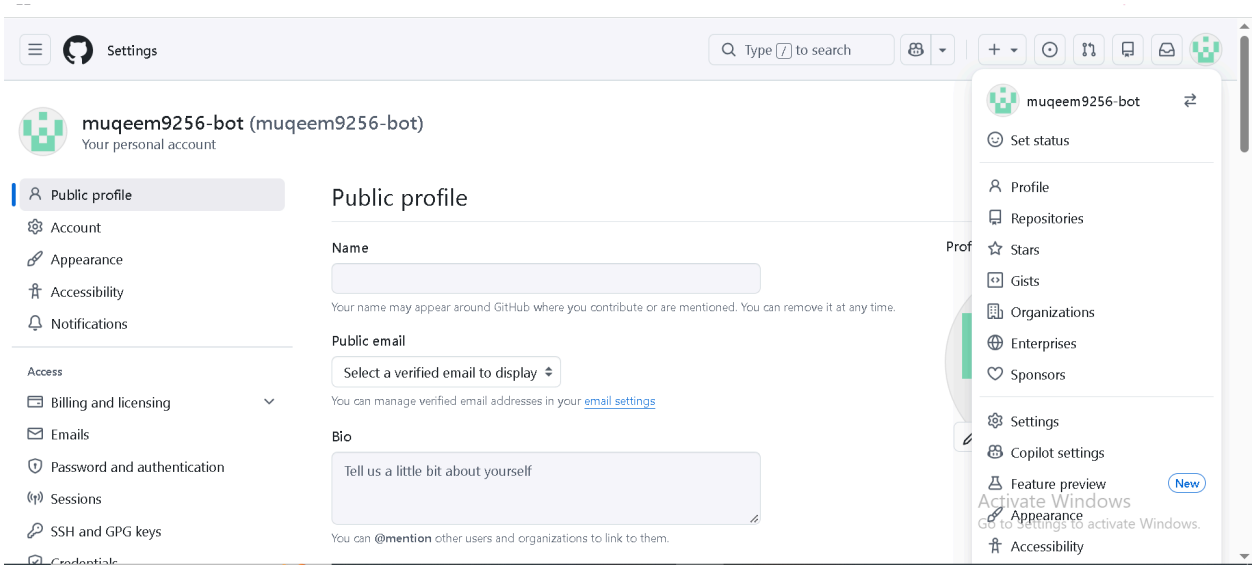
**On your mobile phone, install an authenticator app,**




## Step 2: Open GitHub settings

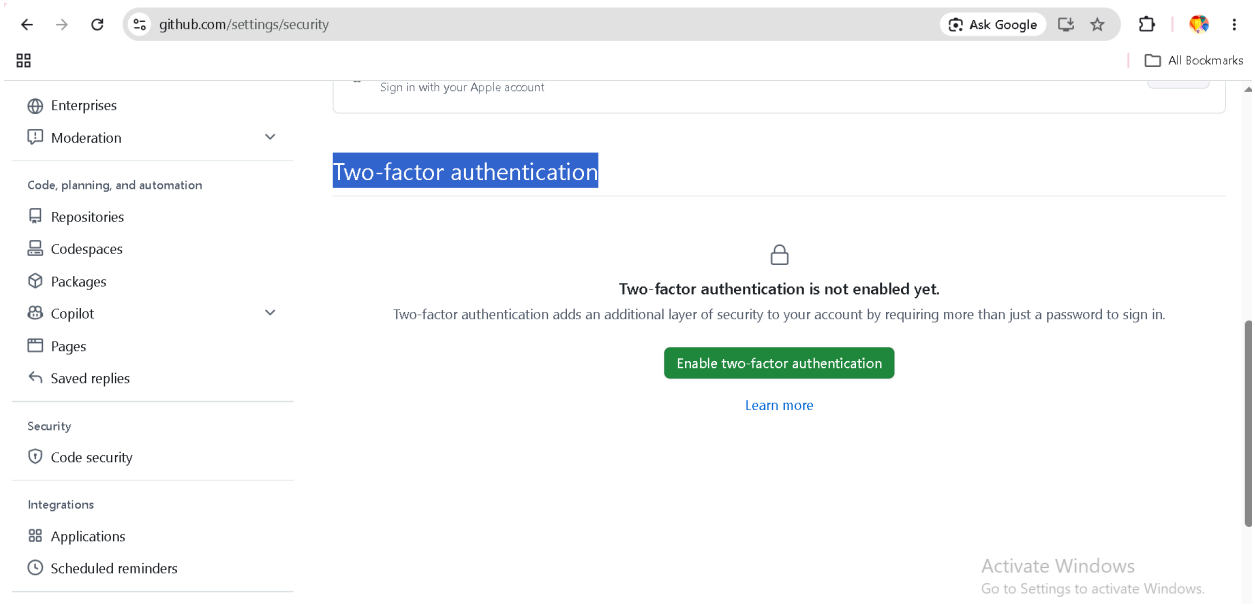
1. **Open github.com on your computer.**
2. **sign in to your account.**
3. **Click your profile picture in the top-right corner.**
4. **Select Settings.**

5. **Click Password and authentication in the sidebar.**

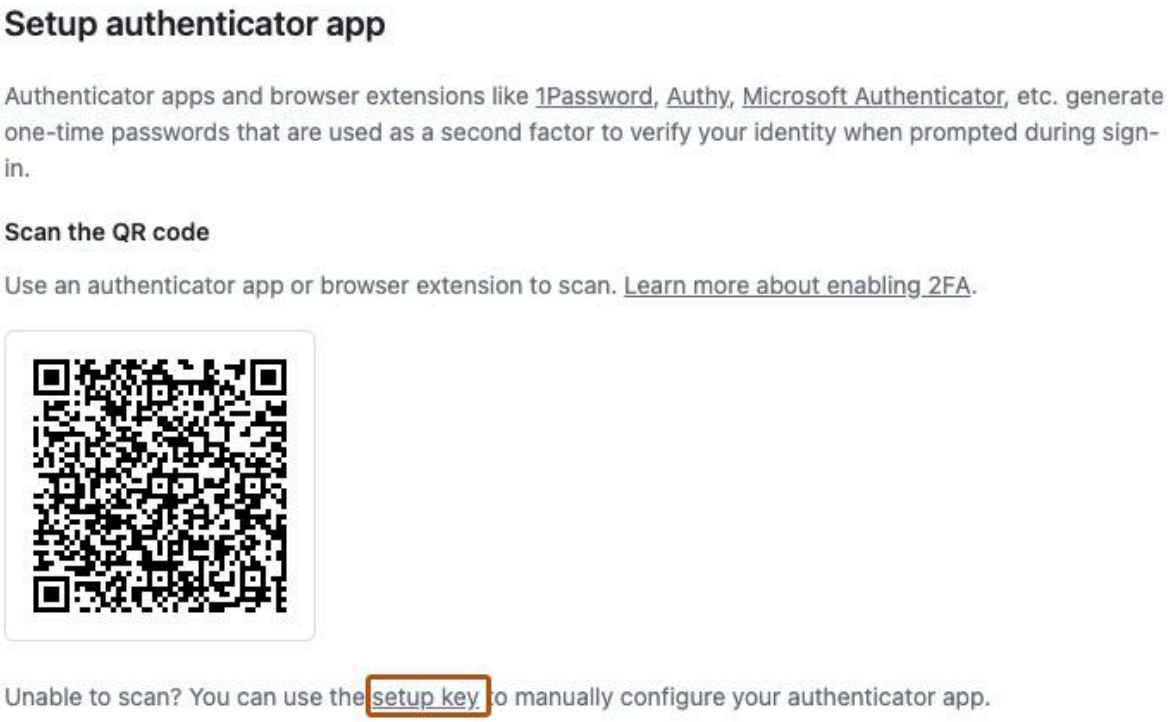



## Step 3: Enable two-factor authentication

1. **Find the Two-factor authentication section.**
2. **Click Enable two-factor authentication.**
3. **Select the authenticator app option if prompted.**




## Step 4: Scan the QR code




**1.GitHub will display a QR code on your computer.**

**2.Open Google Authenticator on your mobile phone.**

**3.Tap the + button and select Scan a QR code.**

**4.Scan the QR code displayed on your computer.**

**5.The app will generate a six-digit verification code.**

## Step 5: Verify the code

- **Enter the six-digit code from your mobile app into GitHub.**
- **Click the verification or continue button shown on the page.**

**Your authenticator app is now connected to your GitHub account.**

## Step 6: Save recovery codes

**GitHub will ask you to save your recovery codes.**

- **Download the recovery codes.**
- **Keep them in a safe place.**
- **Do not share them with anyone.**

**Execute all commands shown in the video.**

**Steps:**

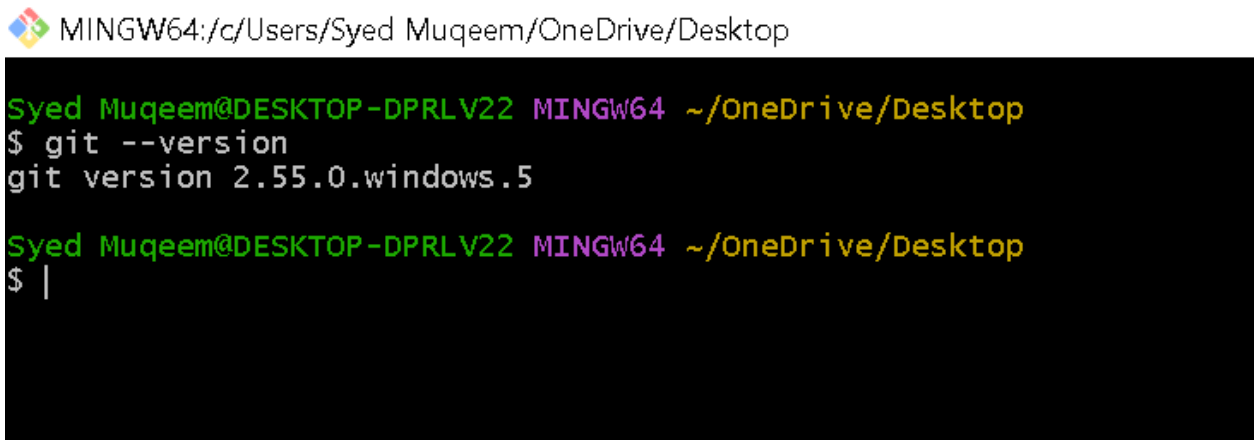
1. **For executing git commands we need to install the git first using the below command**

   **: yum install git**

   **we need to check the version using the below command**

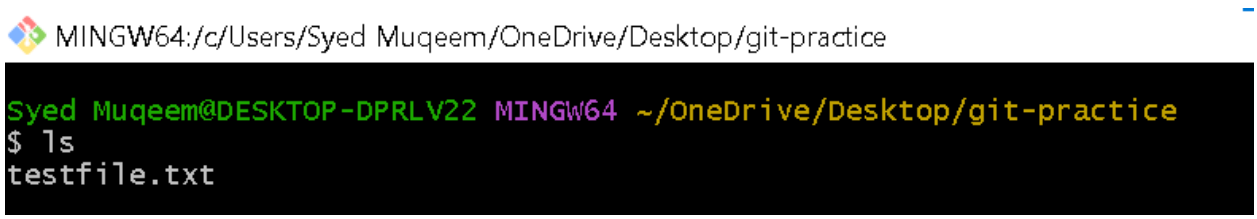
   **: Git - -version**

   **I have already installed so it is showing the version like: 2.55.0**



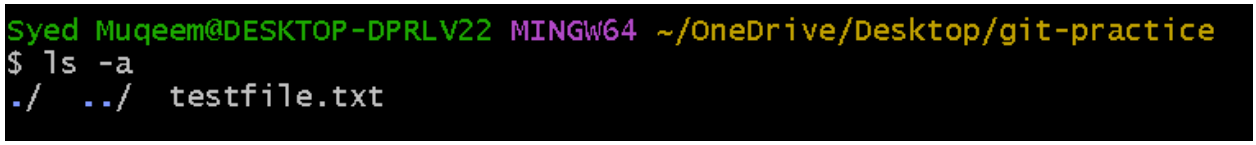

2. **We us ls command to list the files in git**

   **: ls**




3. **We use ls -a command to list the hidden files**

   **: ls -a**



4. **git init we use this command to initialize the tracking of the files**

   **: git init**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice
$ git init
Initialized empty Git repository in C:/Users/Syed Muqeem/OneDrive/Desktop/git-practice/.git/

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$
```

   **# When you run this command a new directory “.git/ ” gets created**

   **# it has all the configurations related to our github like: which project,what should be tracked, what is the branch and who is the master etc.**

   **# Below are the all configuration file undere .git/ directory.**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ cd .git/

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice/.git (GIT_DIR!)
$ ll
total 7
-rw-r--r-- 1 Syed Muqeem 197121  23 Oct 22 15:45 HEAD
-rw-r--r-- 1 Syed Muqeem 197121 130 Oct 22 15:45 config
-rw-r--r-- 1 Syed Muqeem 197121  73 Oct 22 15:45 description
drwxr-xr-x 1 Syed Muqeem 197121   0 Oct 22 15:45 hooks/
drwxr-xr-x 1 Syed Muqeem 197121   0 Oct 22 15:45 info/
drwxr-xr-x 1 Syed Muqeem 197121   0 Oct 22 15:45 objects/
drwxr-xr-x 1 Syed Muqeem 197121   0 Oct 22 15:45 refs/
```

5. **We use the git status command to check the tracking status of the files and directories, after git initialization.**

   **# If you deleted the .git/ directory and you try git status it will not work.**

   **# You need to initialize the tracking using the “ git init ” command.**

   **: git status**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        testfile.txt

nothing added to commit but untracked files present (use "git add" to track)

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$
```

6. **Now we are using the important git commands**

**: git config - -list**

**# is a Git command that displays your current Git configuration settings.**

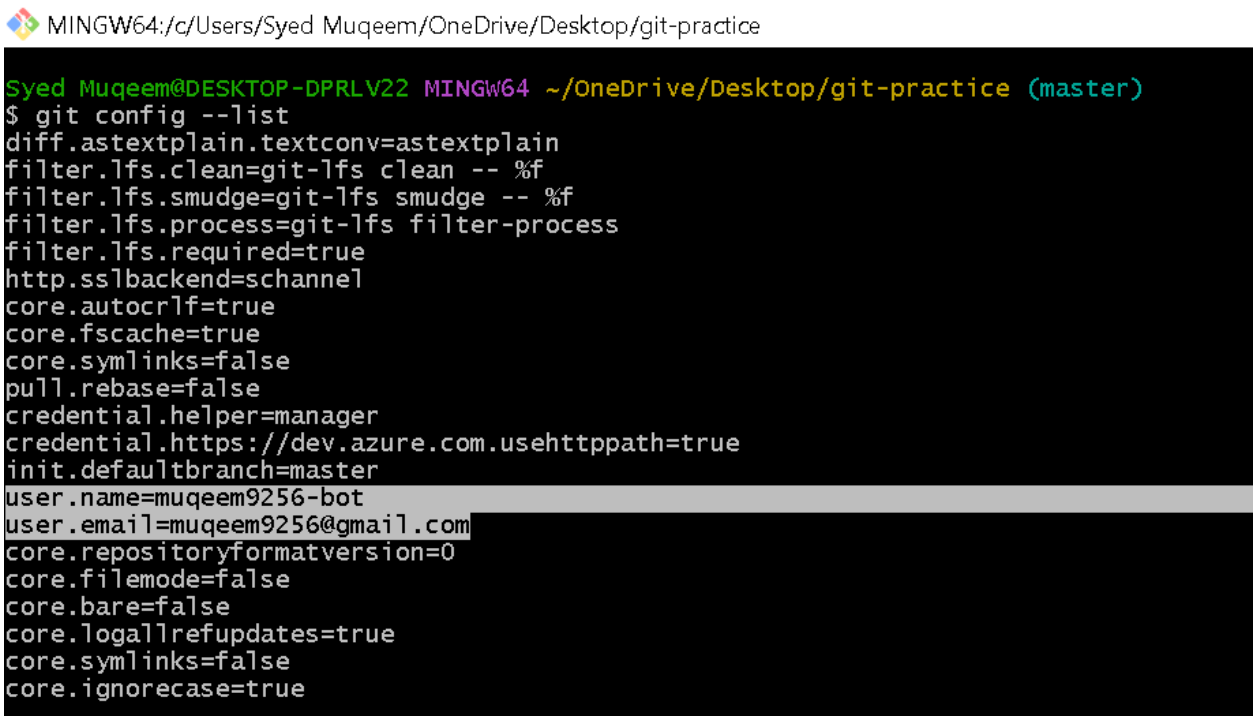
**# without this configuration we cannot push the changes to the repo**

**# it just one time activity , to configure user name use the below command**

**: git config --global user.name "muqeem9256-bot"  ( user name)**

**: git config --global user.email "muqeem9256@gmail.com"  ( user email )**

**: git config - -list**




7. **Now, in order to push your files to the centralized repository.**

   **# we need to move the files from untracking area to the tracking area using the command “git add filename”**

   **: git add testfile.txt**




   **: git status**

   **# Now we can see that the file turned into green color it means has moved to the tracking area.**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   testfile.txt
```

**# if we have multiple files to move to the tracking area we use the command**

**: ” git add . “**

MINGW64:/c/Users/Syed Muqeem/OneDrive/Desktop/git-practice

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   testfile.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        a
        b
        c
        d
        e
        f
        g
        h
        i


Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git add .
```

**# now all the files has been moved from untracking area to the tracking area see below .**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   a
        new file:   b
        new file:   c
        new file:   d
        new file:   e
        new file:   f
        new file:   g
        new file:   h
        new file:   i
        new file:   testfile.txt
```

8. **`git commit -m` is a Git command used to save changes in your local Git repository with a message**

   **: git commit -m “Added some new files”**

MINGW64:/c/Users/Syed Muqeem/OneDrive/Desktop/git-practice

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git commit -m "Added some new files"
[master (root-commit) c3905c4] Added some new files
 10 files changed, 4 insertions(+)
 create mode 100644 a
 create mode 100644 b
 create mode 100644 c
 create mode 100644 d
 create mode 100644 e
 create mode 100644 f
 create mode 100644 g
 create mode 100644 h
 create mode 100644 i
 create mode 100644 testfile.txt

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git status
On branch master
nothing to commit, working tree clean

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ ls
a  b  c  d  e  f  g  h  i  testfile.txt

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$
```

**# ‘Nothing to commit means” now we are ready to push the code from our local machine to the central server.**

9. **`git log` is a Git command used to view the history of previous commits in your repository**

   **Git log**

   **# this c3905c4 is a commit id**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git log
commit c3905c4efab4146d9e4b4c976678d19cadf36d9e (HEAD -> master)
Author: muqeem9256-bot <muqeem9256@gmail.com>
Date:   Thu Oct 22 16:53:34 2026 +0530

    Added some new files
```

   **# created a file “ z “ and added it to the tracking area and committed it with a message and checked the git log below.**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ touch z

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        z

nothing added to commit but untracked files present (use "git add" to track)

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git add z

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git commit -m "added z file"
[master ea44818] added z file
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 z

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git log
commit ea448189c4143dce518f4e8c1106f6e15cb24b5f (HEAD -> master)
Author: muqeem9256-bot <muqeem9256@gmail.com>
Date:   Thu Oct 22 17:05:10 2026 +0530

    added z file

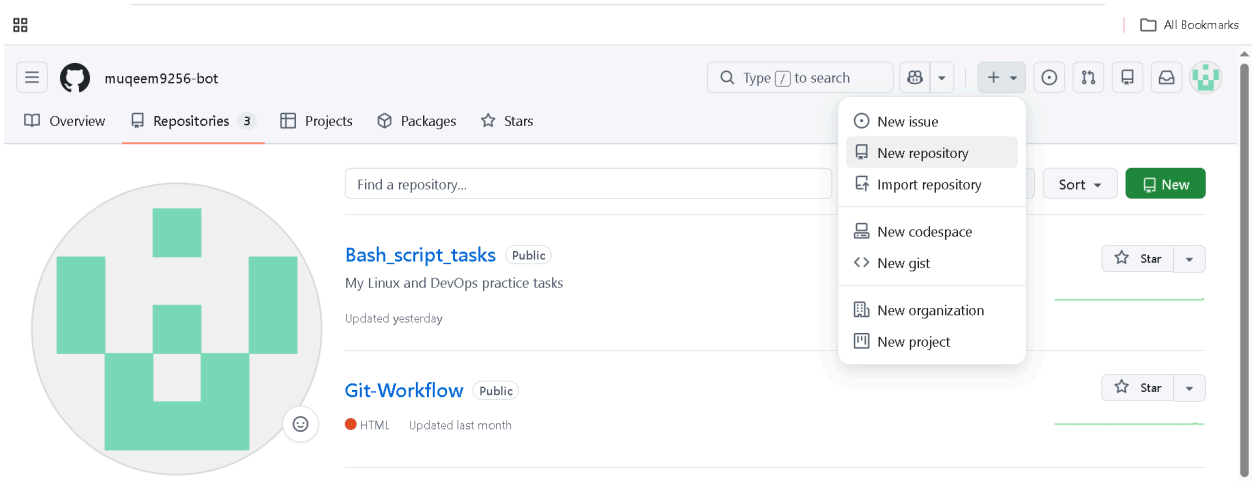
commit c3905c4efab4146d9e4b4c976678d19cadf36d9e
Author: muqeem9256-bot <muqeem9256@gmail.com>
Date:   Thu Oct 22 16:53:34 2026 +0530

    Added some new files
```

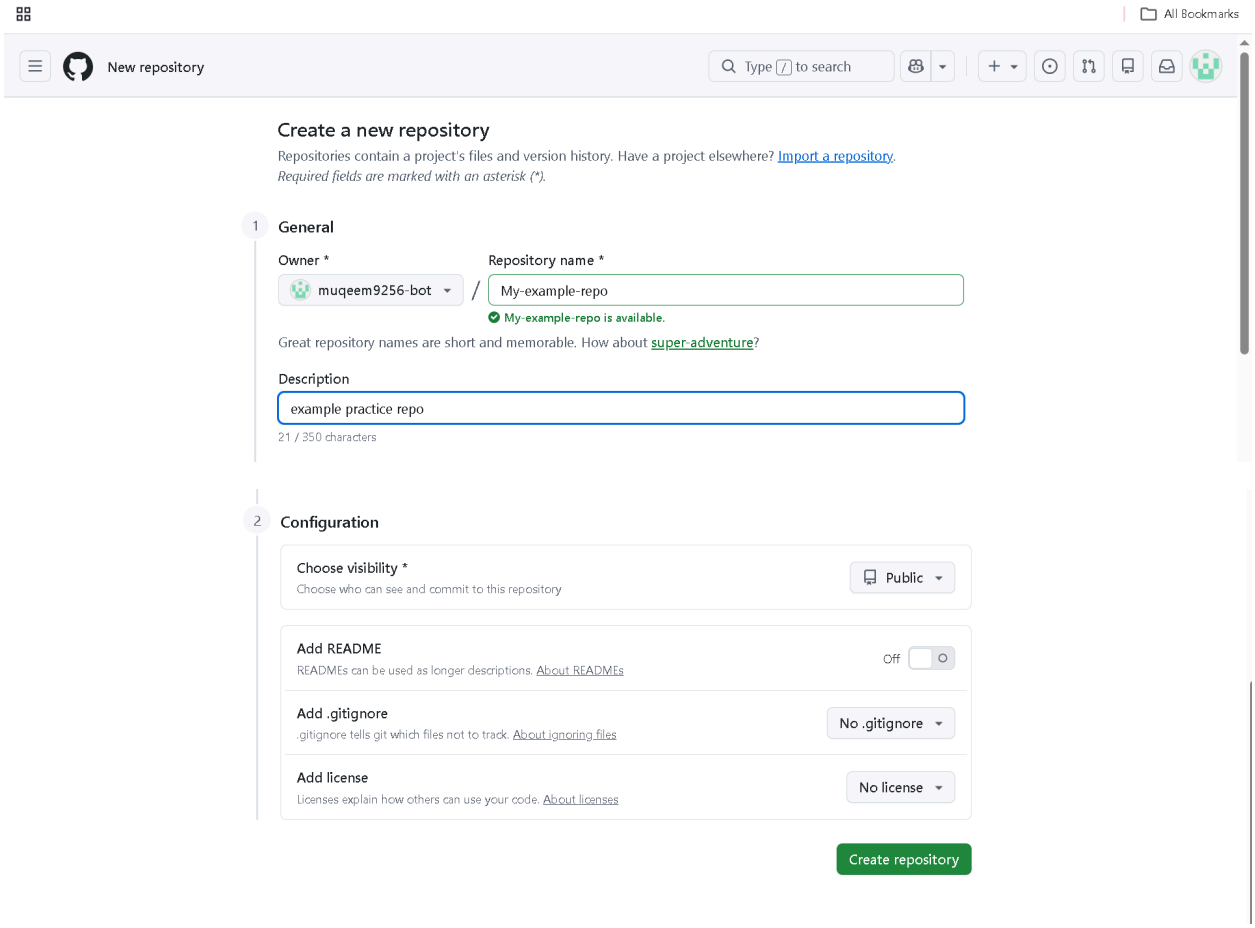
**# now we are ready to push the code from our local machine to the central server.**

**# in order to push your code from your local machine to the github repository**

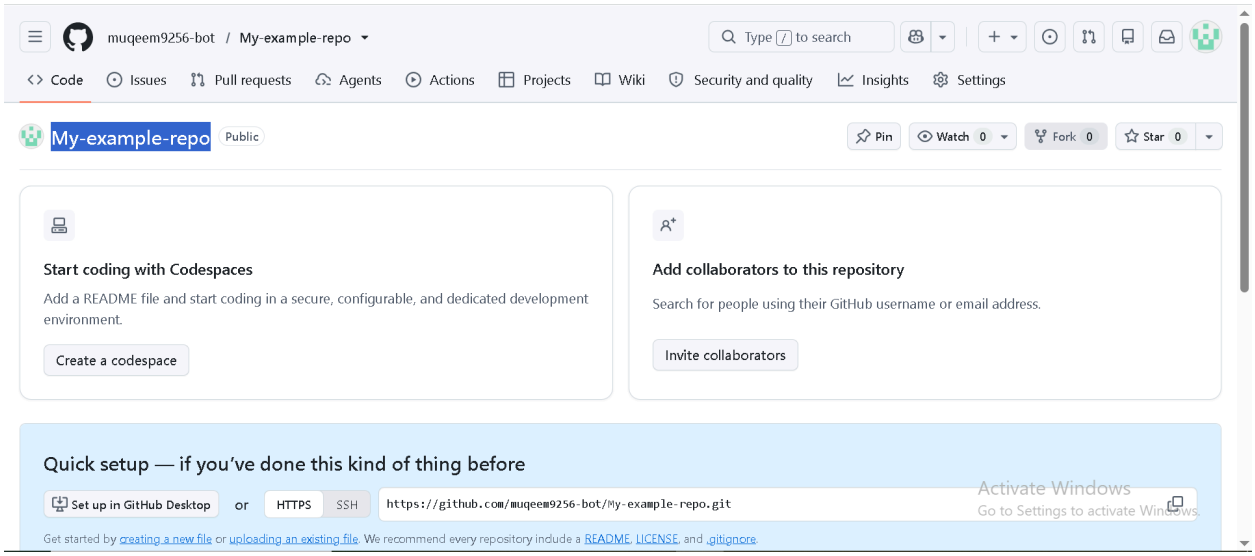
**# you need to go to your github account and create a new repository.**



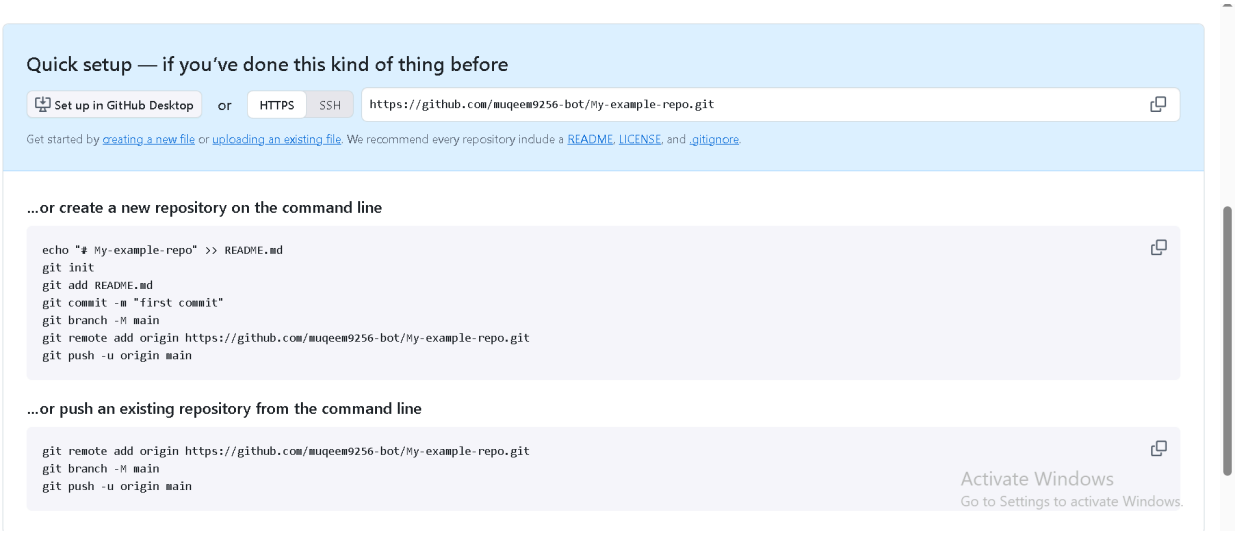

**# click on new repository, you will see this. Provide the details and**




**# Click on create repository. Below is the repository created**




**# the below is the url generated for your repository**




10. **`git remote add origin` is a Git command used to connect your local Git repository to a remote repository on GitHub.**

    **: git remote add origin https://github.com/muqeem9256-bot/My-example-repo.git**

    **# After creating a repository on GitHub, run these commands in your local project folder**

    **# To check the connection  run the below command**

    **: git remote -v**

MINGW64:/c/Users/Syed Muqeem/OneDrive/Desktop/git-practice

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git remote -v
origin  https://github.com/muqeem9256-bot/My-example-repo.git (fetch)
origin  https://github.com/muqeem9256-bot/My-example-repo.git (push)

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$
```

**# Now we can use the git push command to push the code from my local machine**

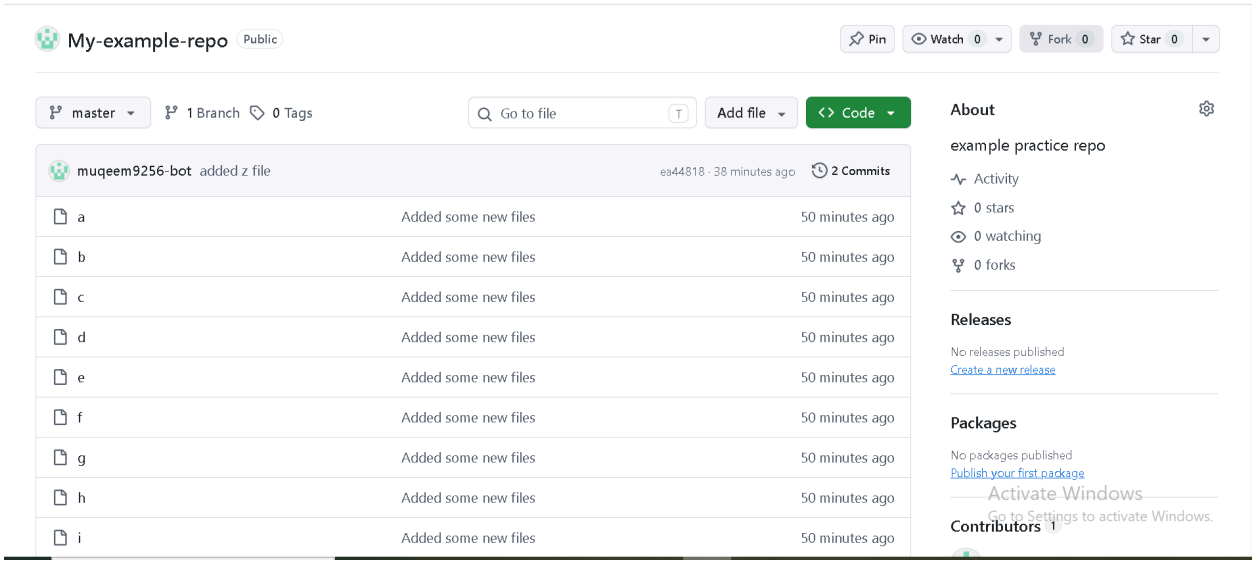
**# but before pushing make sure the connectivity with "git remote -v command"**

**: git push -u origin master**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git push -u origin master
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 497 bytes | 248.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/muqeem9256-bot/My-example-repo.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$
```

**# now go to your github account and refresh, you will see all the files has been pushed to the repository.**




11. **Now I deleted a file ‘a’ from my local machine and want to update the same in the remote repository in the below image.**

    **: rm -rf a**

    **: ls**

    **: git status**

    **: git add .**

    **: git commit -m "Deleted file 'a' "**

    **: git push -u origin master**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ rm -rf a

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ ls
b  c  d  e  f  g  h  i  testfile.txt  z

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    a

no changes added to commit (use "git add" and/or "git commit -a")

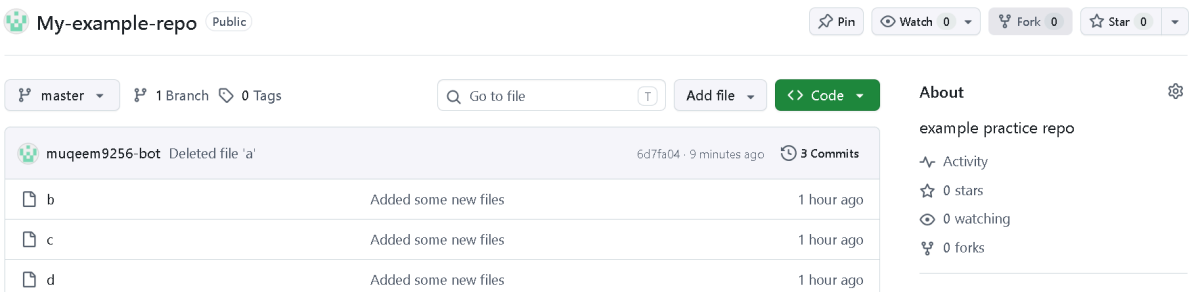
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git add .

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git commit -m "Deleted file 'a'"
[master 6d7fa04] Deleted file 'a'
 1 file changed, 0 insertions(+), 0 deletions(-)
 delete mode 100644 a

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git push -u origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 224 bytes | 224.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/muqeem9256-bot/My-example-repo.git
   ea44818..6d7fa04  master -> master
branch 'master' set up to track 'origin/master'.

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$
```

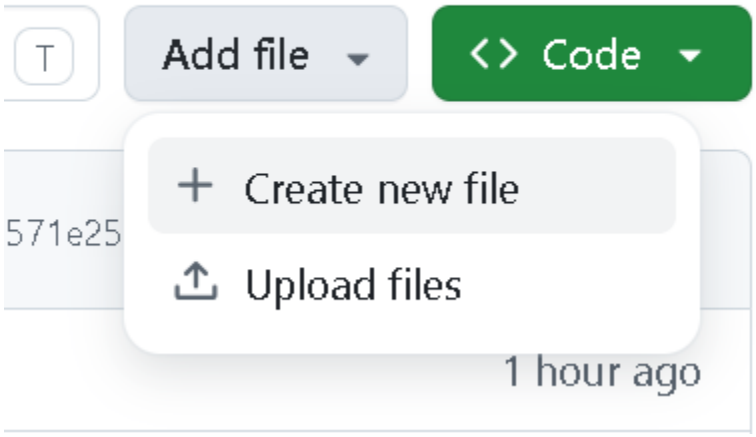
**# now  go to your github account and refresh, you will see the commit message and the updated version of your files after deleting file ‘a’ . see it is showing b, c, d, but not a.**




**# Now i have  added a file in my repository directly from the git hub account**

**: clicking add file drodown**

**: selected create new file**




**: named the file “syed” and added some text in it**

**: and committed manually by a commit message**

**# this new “syed” file will not reflect inn my local machine**

**# And also i cannot push any new code to the remote repository before pulling the remote repo.**

**# example i created a file “abc” and try to push it to my remote repository**

**: touch abc**

**: git add .**

**: git commit -m “added abc file”**

**: git push -u origin master**

MINGW64:/c/Users/Syed Muqeem/OneDrive/Desktop/git-practice

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ touch abc

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git add .

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git commit -m "added abc file "
[master 8de15b3] added abc file
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 abc

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ ls
abc  b  c  d  e  f  g  h  i  testfile.txt  z

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git push -u origin master
To https://github.com/muqeem9256-bot/My-example-repo.git
 ! [rejected]        master -> master (fetch first)
error: failed to push some refs to 'https://github.com/muqeem9256-bot/My-example-repo.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

**# for that we need to pull the remote repo updated files to my local machine using : git pull**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git pull
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 937 bytes | 37.00 KiB/s, done.
From https://github.com/muqeem9256-bot/My-example-repo
   6d7fa04..b571e25  master     -> origin/master
Merge made by the 'ort' strategy.
 syed | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 syed

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ ls
abc  b  c  d  e  f  g  h  i  syed  testfile.txt  z

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$
```

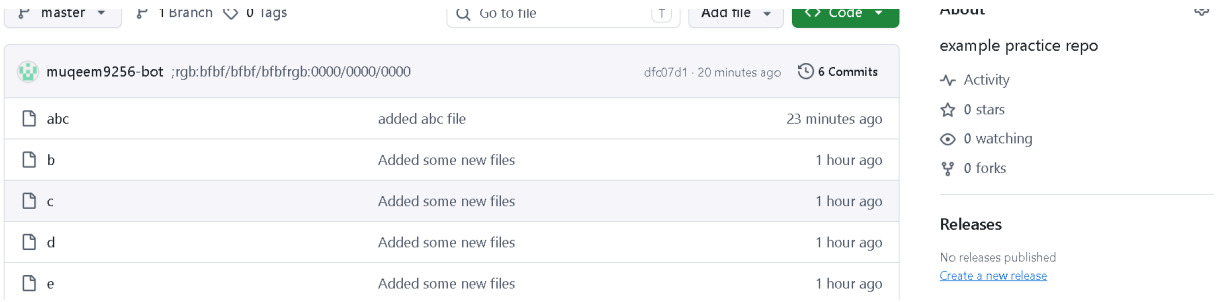
**# you can see here the file name “syed ” we can see in the local machine**

**# now we can push our new added file “abc” to our remote repository**

**: git push -u origin master**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git push -u origin master
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 475 bytes | 237.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/muqeem9256-bot/My-example-repo.git
   b571e25..dfc07d1  master -> master
branch 'master' set up to track 'origin/master'.
```

**# now if i go and refresh the github account then i can see the “abc” file has been added.**




## 12. Git clone <repo link>

**#`git clone` is a Git command used to download a copy of a repository from GitHub to your computer, including its files and commit history**

**: git clone git@github.com:muqeem9256-bot/My-example-repo.git**

MINGW64:/c/Users/Syed Muqeem/OneDrive/Desktop/git-practice/My-example-repo

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ git clone git@github.com:muqeem9256-bot/My-example-repo.git
Cloning into 'My-example-repo'...
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 15 (delta 5), reused 12 (delta 4), pack-reused 0 (from 0)
Receiving objects: 100% (15/15), done.
Resolving deltas: 100% (5/5), done.

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ ls
My-example-repo/  abc  b  c  d  e  f  g  h  i  syed  testfile.txt  z

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice (master)
$ cd My-example-repo/

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice/My-example-repo (master)
$ ls
abc  b  c  d  e  f  g  h  i  syed  testfile.txt  z

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice/My-example-repo (master)
$
```

**13.`git revert` is a Git command used to undo the changes made by a previous commit without deleting the commit history.**

**: git revert <commit id>**

**# to get the commit id we use the command**

**: git log -4  (-4 is las 4 commit logs will display)**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice/My-example-repo (master)
$ git log -4
commit dfc07d17e2fa0acebe8402f8a835bfeb1790d729 (HEAD -> master, origin/master, origin/HEAD)
Merge: 8de15b3 b571e25
Author: muqeem9256-bot <muqeem9256@gmail.com>
Date:   Thu Oct 22 18:10:01 2026 +0530

    ;rgb:bfbf/bfbf/bfbfrgb:0000/0000/0000

commit 8de15b3ad5c6e7981631097ca84a6e1908c3d4be
Author: muqeem9256-bot <muqeem9256@gmail.com>
Date:   Thu Oct 22 18:06:44 2026 +0530

    added abc file

commit b571e25a05cf9a8ec50e691615dcc511e89f8e5b
Author: muqeem9256-bot <muqeem9256@gmail.com>
Date:   Thu Oct 22 18:03:20 2026 +0530

    file on central repo

commit 6d7fa04b9147a9e330c26f39f766c50d48f6f6af
Author: muqeem9256-bot <muqeem9256@gmail.com>
Date:   Thu Oct 22 17:50:34 2026 +0530

    Deleted file 'a'

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice/My-example-repo (master)
$
```

**# now we can copy the commit id 6d7fa0 in the command**

**: git rvert 6d7fa0**

**# now you can see the file ‘a’ is reflecting when i do ls.**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice/My-example-repo (master)
$ git revert 6d7fa0
[master e2cc740] ;rgb:bfbf/bfbf/bfbfrgb:0000/0000/0000
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 a

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice/My-example-repo (master)
$ ls
a  abc  b  c  d  e  f  g  h  i  syed  testfile.txt  z

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice/My-example-repo (master)
$
```

**14. `git reset` is a Git command used to move the current branch to an earlier commit. Depending on the option you use, it can also change what happens to your staged and working files.**

**# I have created a file “alphabet”**

**: touch alphabet**

**: git add .**

**: git status**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice/My-example-repo (master)
$ touch alphabet

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice/My-example-repo (master)
$ git status
```

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice/My-example-repo (master)
$ git add .

Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice/My-example-repo (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   alphabet
```

**# it is moved to tracking area**

**#now i want that file back in the untracking area, then this command we use**

**: git reset**

**# now you can see the file is under the untracking area.**

```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/OneDrive/Desktop/git-practice/My-example-repo (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        alphabet

nothing added to commit but untracked files present (use "git add" to track)
```

**#COMPLETED.**